Software Requirements Engineering

Lecture 4: Software Requirements Elicitation

Review

1. Who is the user?
2. Types and levels of requirements
   - business, user, and functional
   - come from different sources at different times during the project, have different audiences and purposes, and need to be documented in different ways
3. Functional and non-functional requirements

Outline

1. Requirement Elicitation Process
2. Requirement Elicitation Techniques

Requirement Elicitation

- The heart of requirements development is elicitation, the process of identifying the needs and constraints of the various stakeholders for a software system.
- Elicitation is a collaborative and analytical process that includes activities to collect, discover, extract, and define requirements.
- Elicitation is used to discover business, user, functional, and nonfunctional requirements, along with other types of information.

Requirement Elicitation

- Find out and understand the problems to be solved
- Find out and understand the goal and objectives (in the form of systems/services needed)
- Find out the required performance of the solution
- Find out constraints, such as hardware and other, on the solution
- …



Requirement Elicitation

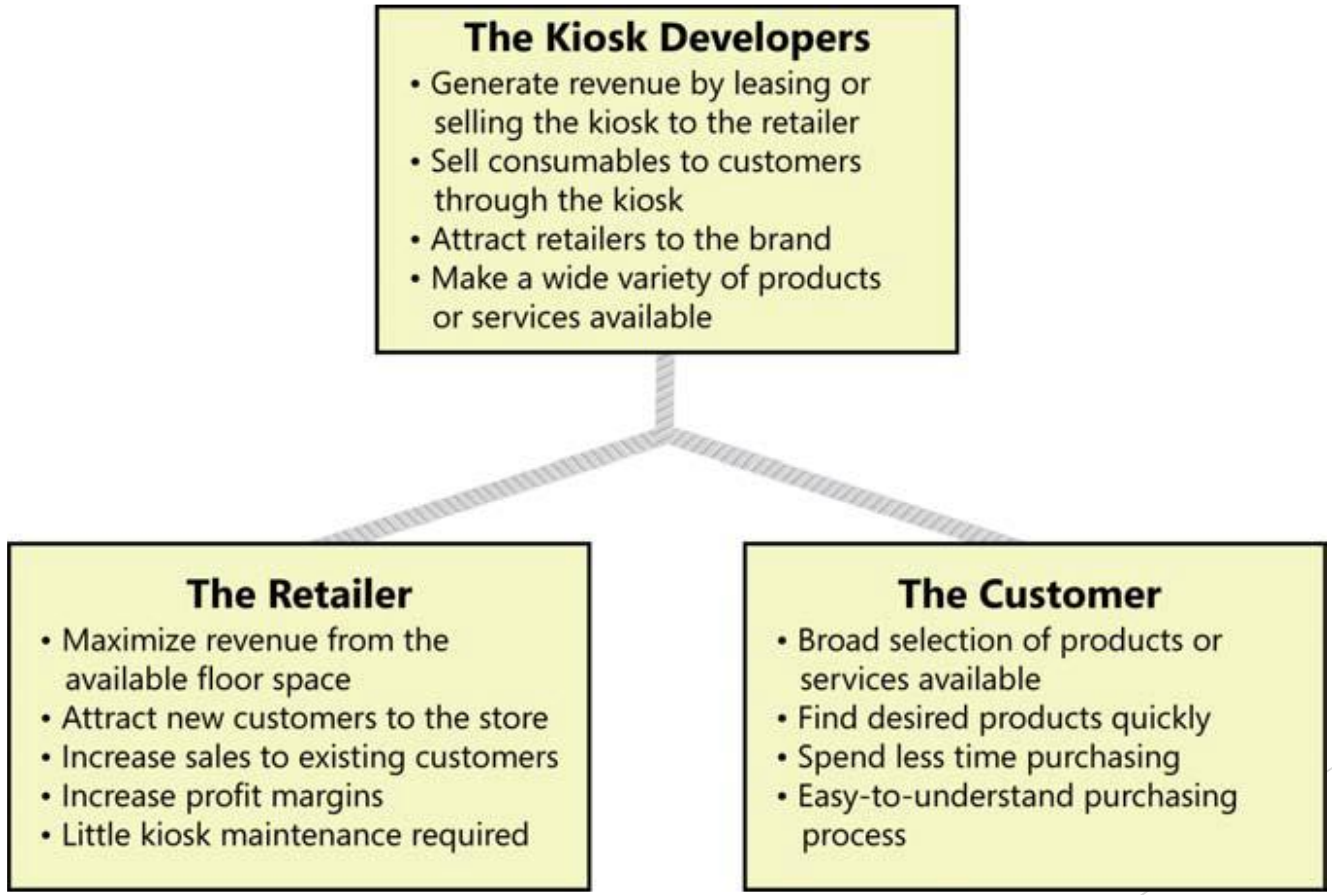
- Define product vision and project scope
- Identify user classes and their characteristics
- Select a product champion for each user class
- Conduct focus groups with typical users
- Work with user representatives to identify user requirements
- Identify system events and responses
- Hold elicitation interviews and facilitated elicitation workshops
- Observe users performing their jobs
- Distribute questionnaires
- Perform document analysis
- Examine problem reports of current systems for requirement ideas
- Reuse existing requirements

Product vision and project scope

- Two core elements of the business requirements are the vision and the scope.
- The product vision succinctly describes the ultimate product that will achieve the business objectives.
- The vision describes what the product is about and what it ultimately could become.
- The product vision ensures that we all know where we are hoping to go eventually.
- The project scope ensures that we are all talking about the same thing for the immediate project or iteration.

Conflicting business requirements

- Business requirements collected from multiple sources might conflict. Consider a kiosk that will be used by a retail store's customers.



Vision and scope document

- The vision and scope document collects the business requirements into a single deliverable that sets the stage for the subsequent development work.
- A template is as shown.
- The vision and scope document only defines the scope at a high level;

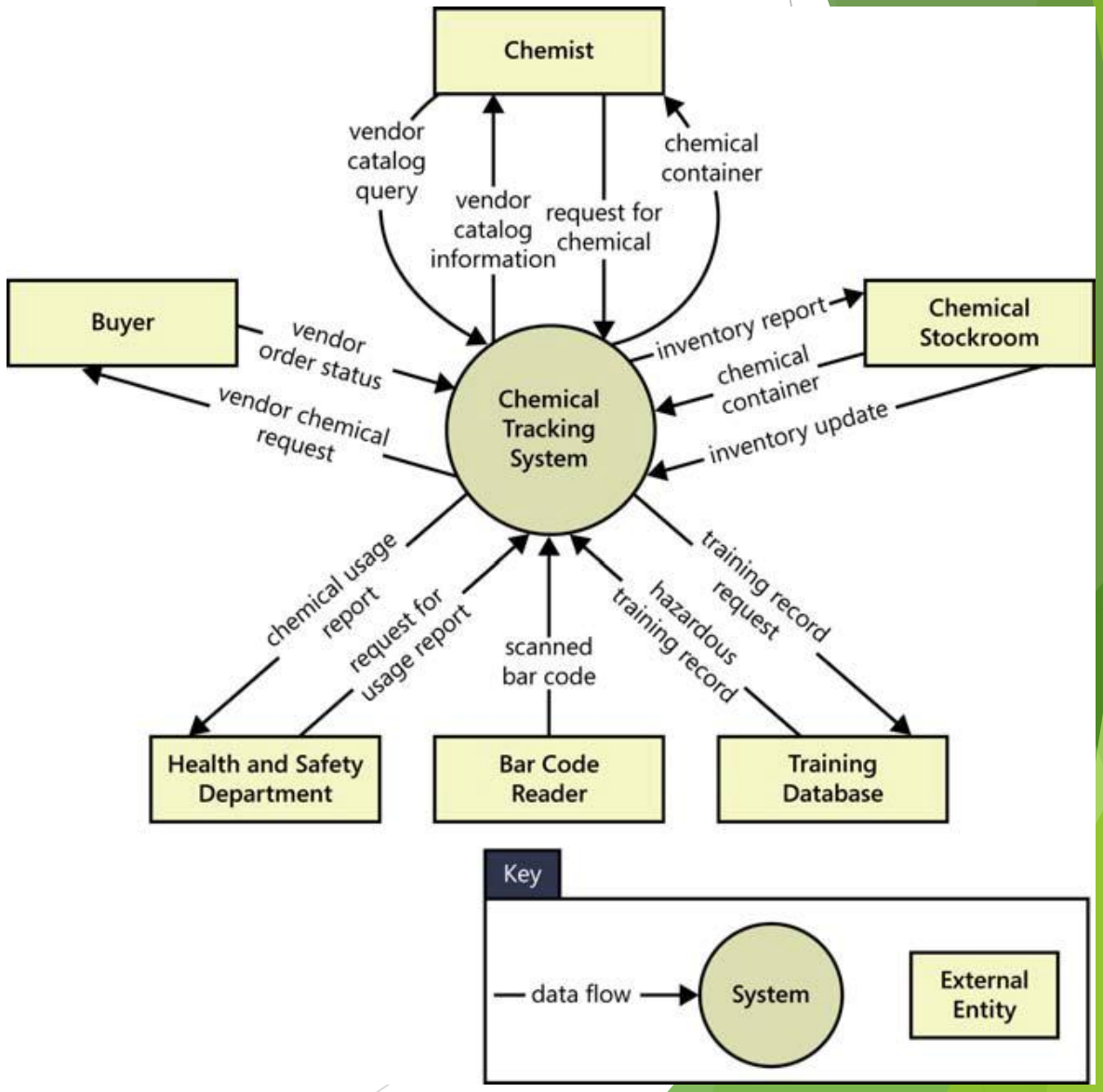
1. Business requirements
- 1.1 Background
- 1.2 Business opportunity
- 1.3 Business objectives
- 1.4 Success metrics
- 1.5 Vision statement
- 1.6 Business risks
- 1.7 Business assumptions and dependencies

2. Scope and limitations
- 2.1 Major features
- 2.2 Scope of initial release
- 2.3 Scope of subsequent releases
- 2.4 Limitations and exclusions

3. Business context
- 3.1 Stakeholder profiles
- 3.2 Project priorities
- 3.3 Deployment considerations

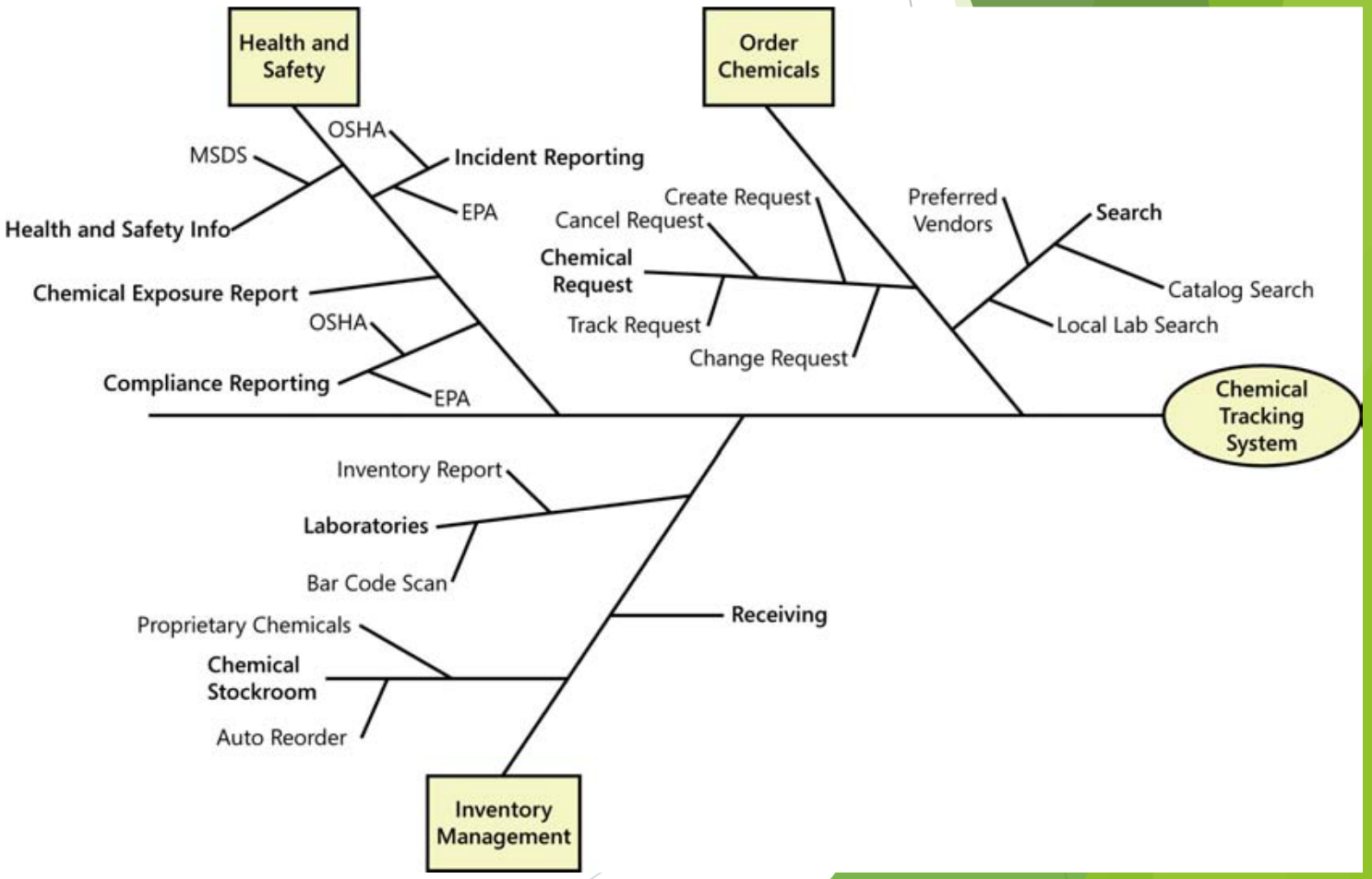
Scope representation techniques

- Context diagram
- The context diagram identifies external entities (also called terminators) outside the system that interface to it in some way, as well as data, control, and material flows between the terminators and the system.



Scope representation techniques

- Feature tree
- A feature tree is a visual depiction of the product's features organized in logical groups
- hierarchically subdividing each feature into further levels of detail.
- The feature tree provides a concise view of all of the features planned for a project

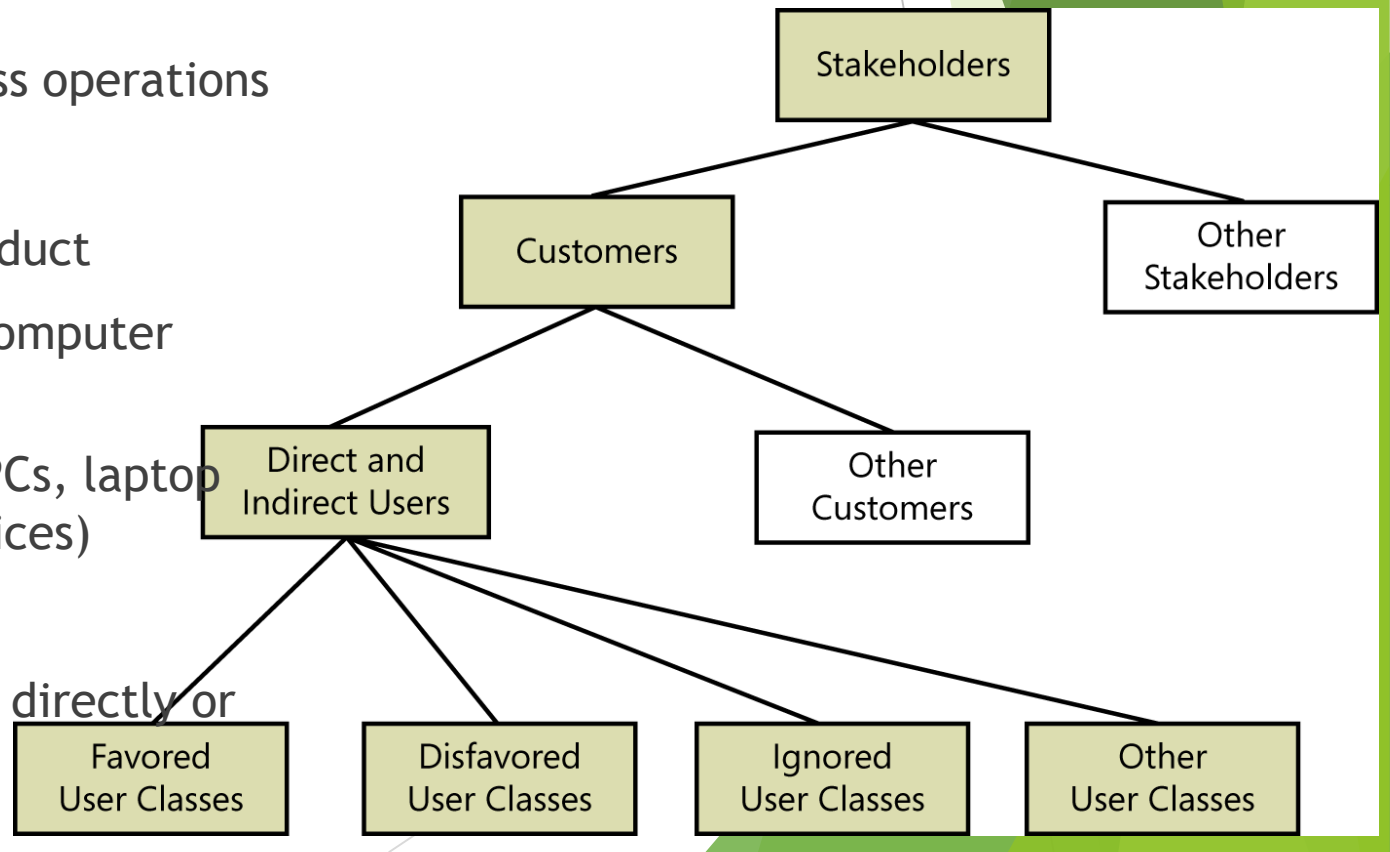


Closing the Expectation Gap

- The success in software requirements, and hence in software development, depends on getting the voice of the user close to the ear of the developer. To find the voice of the user, you need to take the following steps:
  - Identify the different classes of users for your product.
  - Select and work with individuals who represent each user class and other stakeholder groups.
  - Agree on who the requirements decision makers are for your project.
- Customer involvement is the best way to avoid the expectation gap between the product that customers expect to receive and what developers build.

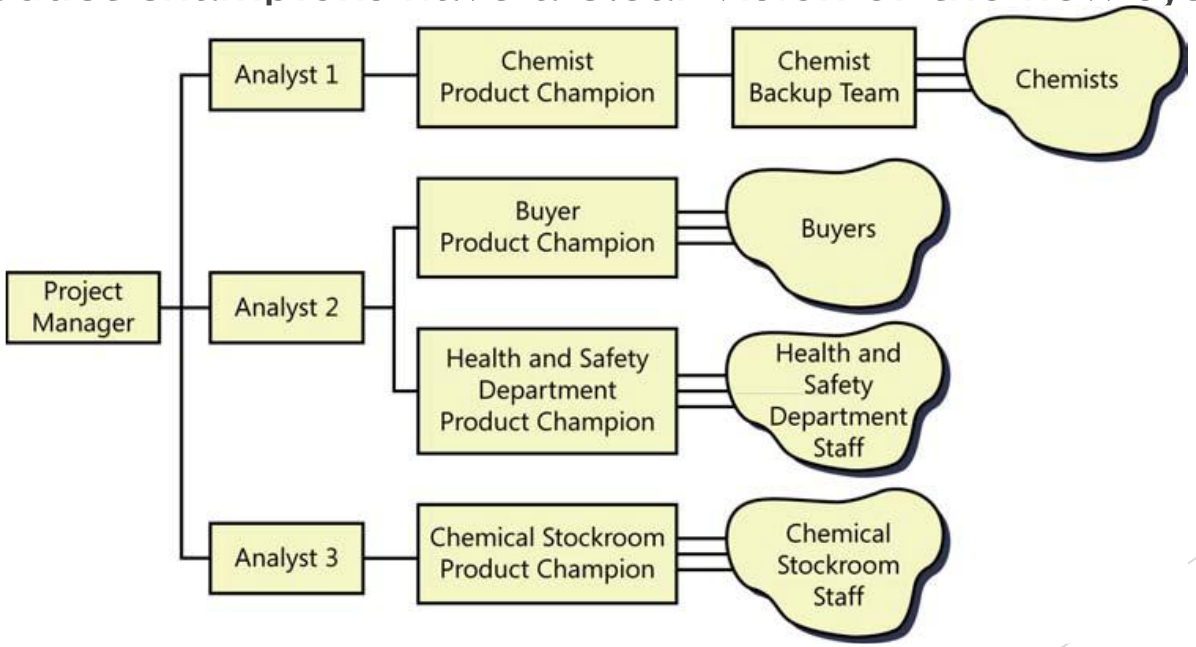
Classifying users

- We can group users into a number of distinct user classes based on these sorts of differences:
  - Their access privilege or security levels (such as ordinary user, guest user, administrator)
  - The tasks they perform during their business operations
  - The features they use
  - The frequency with which they use the product
  - Their application domain experience and computer systems expertise
  - The platforms they will be using (desktop PCs, laptop PCs, tablets, smartphones, specialized devices)
  - Their native language
  - Whether they will interact with the system directly or indirectly

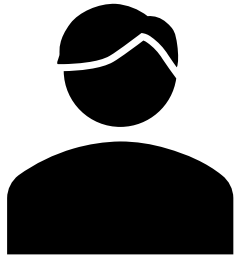


The product champion

- Each product champion serves as the primary interface between members of a single user class and the project's business analyst.
- Product champions gather requirements from other members of the user classes they represent and reconcile inconsistencies.
- The best product champions have a clear vision of the new system.



Introduction



Good morning, Maria. I'm Phil, the business analyst for the new employee information system we're going to build for you. Thanks for agreeing to be the product champion for this project. Your input will help us a lot. So, can you tell me what you want?

Hmmm, what do I want? I hardly know where to start. The new system should be a lot faster than the old one. And you know how the old system crashes if an employee has a really long name and we have to call the help desk and ask them to enter the name for us? The new system should take long names without crashing. Also, a new law says we can't use Social Security numbers for employee IDs anymore…



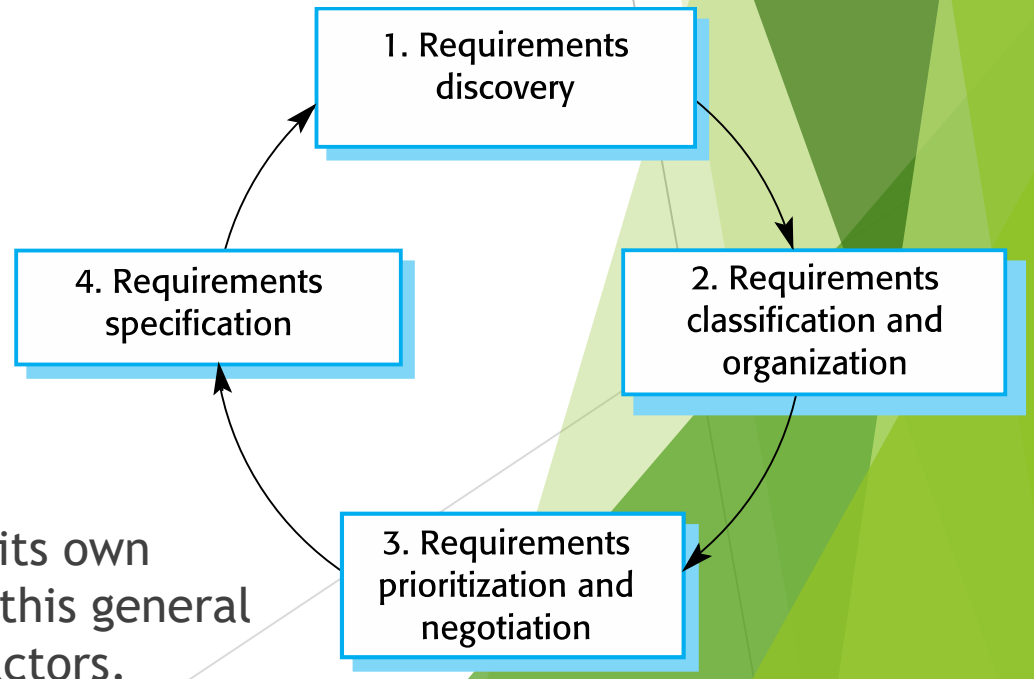
Phil dutifully wrote down everything Maria said, but his head was spinning. Maria's desires were so scattered that he wasn't sure he was getting all her requirements. He had no idea if Maria's needs aligned with the project's business objectives. And he didn't know exactly what to do with all these bits of information. Phil wasn't sure what to ask next.

Requirements elicitation and analysis

- Sometimes called requirements elicitation or requirements discovery.
- Involves technical staff working with customers to find out about the application domain, the services that the system should provide and the system's operational constraints.
- May involve end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc. These are called *stakeholders*.
- The output of requirements development is a common understanding of the needs held by the diverse project stakeholders.

Requirements elicitation

- Software engineers work with a range of system stakeholders to find out about the application domain, the services that the system should provide, the required system performance, hardware constraints, other systems, etc.
- Stages include:
  - Requirements discovery,
  - Requirements classification and organization,
  - Requirements prioritization and negotiation,
  - Requirements specification.



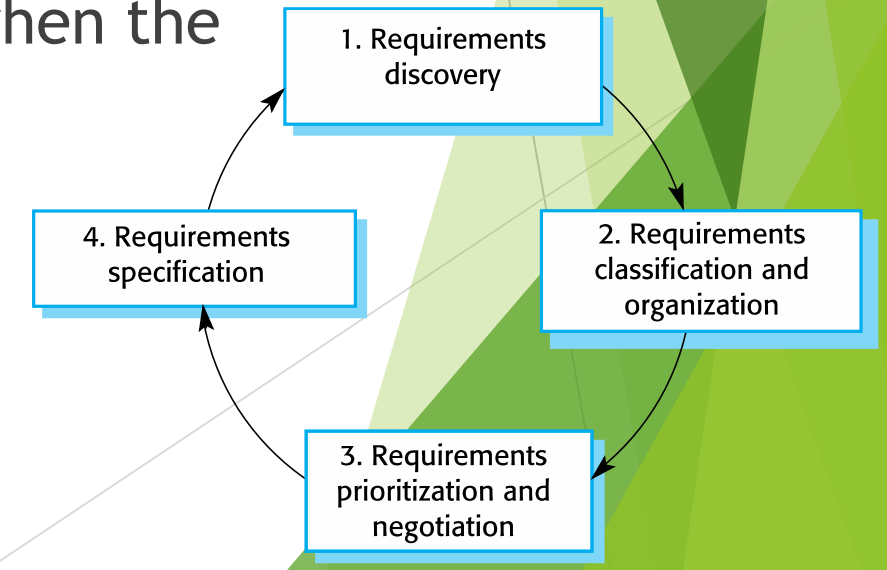
Each organization will have its own version and instantiation of this general model depending on local factors.

Process activities

- Requirements discovery
  - Interacting with stakeholders to discover their requirements. Domain requirements are also discovered at this stage.
- Requirements classification and organisation
  - Groups related requirements and organises them into coherent clusters.
  - The most common way of grouping requirements is to use a model of the system architecture to identify sub-systems and to associated requirements with each sub-system.
- Requirements prioritisation and negotiation
  - Prioritising requirements and resolving requirements conflicts.
- Requirements specification
  - Requirements are documented and input into the next round of the spiral.

Process activities

- The requirements elicitation and analysis is an iterative process with continual feedback from each activity to other activities.
- The process cycle starts with requirements discovery and ends with the requirements documentation.
- The analyst's understanding of the requirements improves with each round of the cycle. The cycle ends when the requirements document is complete.



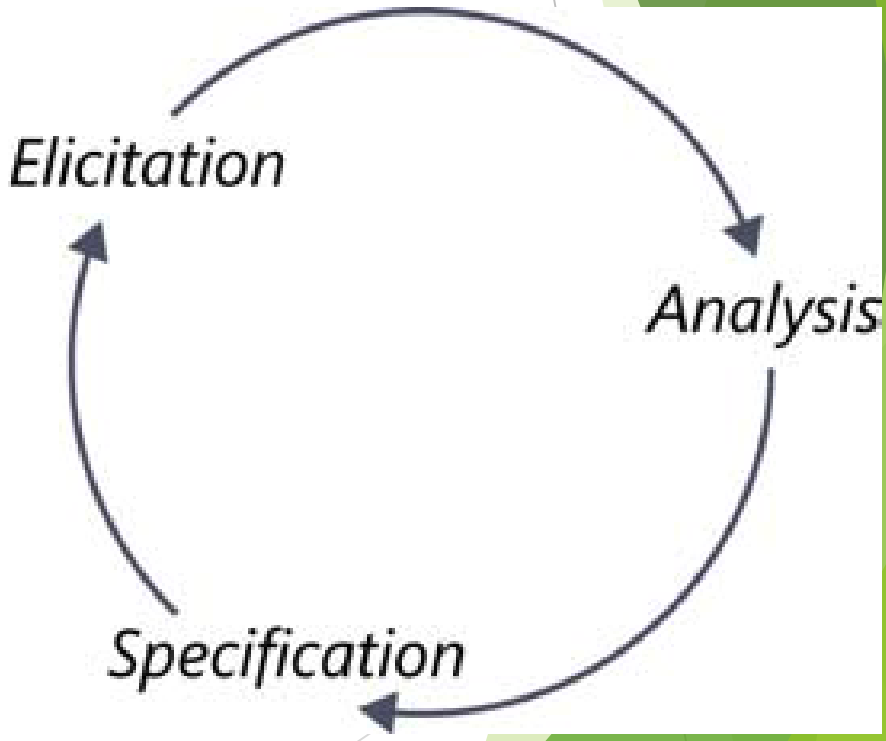
Problems of requirements elicitation

- Eliciting and understanding requirements from system stakeholders is a difficult process for several reasons:

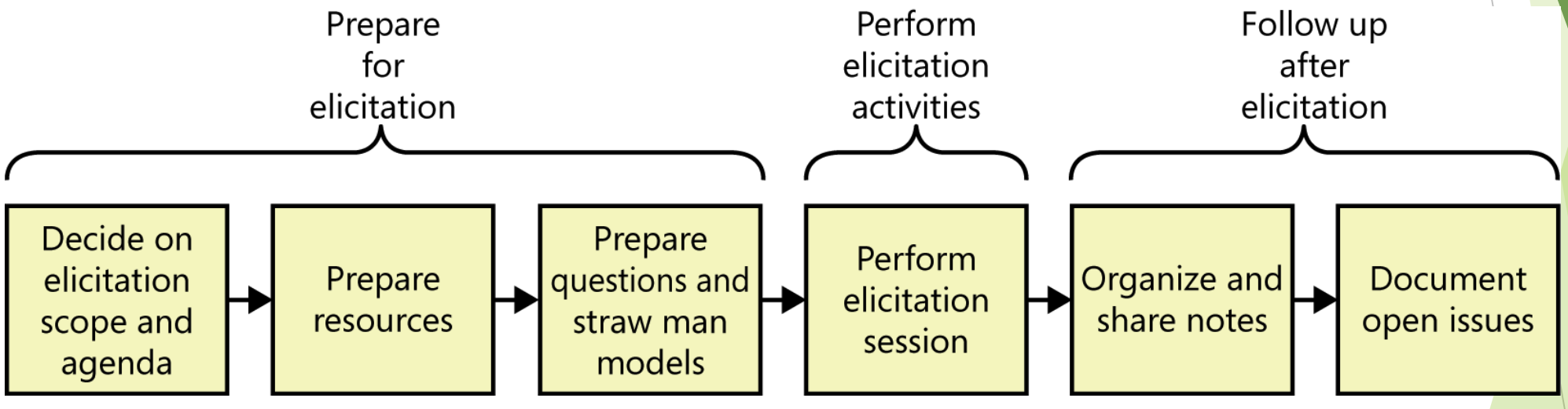
1. Stakeholders don’t know what they really want.
2. Stakeholders express requirements in their own terms.
3. Different stakeholders may have conflicting requirements.
4. Organisational and political factors may influence the system requirements.
5. The requirements change during the analysis process. New stakeholders may emerge and the business environment may change.

Software Requirement Development

- The nature of requirements development is cyclic.
- Don’t expect to just hold a couple of elicitation workshops and then declare victory and move on.



Requirement Elicitation



Activities for a single requirements elicitation session.

Requirements elicitation techniques

- Numerous elicitation techniques can be employed on software projects.
- In fact, no project team should expect to use only one elicitation technique.
- There are always many types of information to be discovered, and different stakeholders will prefer different approaches.
- Elicitation techniques include both facilitated activities, in which you interact with stakeholders to elicit requirements, and independent activities, in which you work on your own to discover information.
- Facilitated activities primarily focus on discovering business and user requirements.
- Most projects will use a combination of both facilitated and independent elicitation activities.

Requirements elicitation techniques

- Interviews
- Workshops
- Focus groups
- Observations
- Questionnaires
- System interface analysis
- User interface analysis
- Document analysis

Interviewing

- Formal or informal interviews with stakeholders are part of most RE processes.
- Types of interview
  - Closed interviews based on pre-determined list of questions
  - Open interviews where various issues are explored with stakeholders.



Interviewing

- Requires preparation and good communication management
  - Achieve interview objectives without preventing the exploration of promising leads
- Interview as many stakeholders as possible
  - Not just clients and users.
- Ask problem-oriented questions
  - Ask about specific details, but…
  - Detailed and solution-specific questions may miss the stakeholder's real requirements.
  - E.g. Would you like Word 2010, Excel 2010 or both?
  - Vs.
  - Would you like to do word processing, computations or both?

Interviewing

- Three main objectives:
  - Record information to be used as input to requirements analysis and modeling
  - Discover information from interviewee accurately and efficiently
  - Reassure interviewee that his/her understanding of the topic has been explored, listened to and valued
- Process consists of four important steps
- Use some strategies for questioning

Interviews in practice

- Normally a mix of closed and open-ended interviewing.
- Interviews are good for getting an overall understanding of what stakeholders do and how they might interact with the system.
- Interviewers need to be open-minded without pre-conceived ideas of what the system should do
- You need to prompt the use to talk about the system by suggesting requirements rather than simply asking them what they want.

Problems with interviews

- Application specialists may use language to describe their work that isn't easy for the requirements engineer to understand.
- Interviews are not good for understanding domain requirements
  - Requirements engineers cannot understand specific domain terminology;
  - Some domain knowledge is so familiar that people find it hard to articulate or think that it isn't worth articulating.

Effective interviewers

- They are open-minded, avoid pre-conceived ideas about the requirements, and are willing to listen to stake holders. If the stakeholder comes up with surprising requirements, then they are willing to change their mind about the system.
- They prompt the interviewee to get discussions going using a springboard question, a requirements proposal, or by working together on a prototype system.

Interviewing

- Important to plan and prepare interviews
- Set goals and objectives for the interview
- Acquire background knowledge of the subject matter to conduct an effective interview
- Prepare questions in advance, by subject
- Organize the environment for conducting an effective interview
  - Determine how the elicitation notes will be taken(manually, audio, video, by whom…)

Interviewing

- Suggestions on conducting interviewing
  - Establish rapport
  - Stay in scope
  - Prepare questions and straw man models ahead of time
  - Suggest ideas
  - Listen actively

Interview Report Template

Topic: Interviewer: Date: Place: Interviewee: Other participants: Objectives:	
Question:	Feedback:
1.1 The necessity of generating daily report	Weekly report is enough
1.2 xxx	xxx
Target of the next meeting:	

Workshops

- Workshops encourage stakeholder collaboration in defining requirements.
- A requirements workshop is a structured meeting to define, create, refine, and reach closure on deliverables
- Workshops are facilitated sessions with multiple stakeholders and formal roles

Workshops

- Tips for conducting effective elicitation workshops
  - Establish and enforce ground rules
  - Fill all of the team roles
  - Plan an agenda
  - Stay in scope
  - Use parking lots to capture items for later consideration
  - Timebox discussions
  - Keep the team small but include the right stakeholders
  - Keep everyone engaged

Focus Groups

- A focus group is a representative group of users who convene in a facilitated elicitation activity to generate input and ideas on a product's functional and quality requirements.
- Focus groups are useful for exploring users' attitudes, impressions, preferences, and needs.
- Focus groups must be facilitated.
- You will need to keep them on topic, but without influencing the opinions being expressed.

Focus Groups



Conducting a Requirements-Gathering Meeting

The scene: A meeting room. The first requirements-gathering meeting is in progress.

The players: Jamie Lazar, software team member; Vinod Raman, software team member; Ed Robbins, software team member; Doug Miller, software engineering manager; three members of marketing; a product engineering representative; and a facilitator.

The conversation:

Facilitator (pointing at whiteboard): So that's the current list of objects and services for the home security function.

Marketing person: That about covers it from our point of view.

Vinod: Didn't someone mention that they wanted all *SafeHome* functionality to be accessible via the Internet? That would include the home security function, no?

Marketing person: Yes, that's right . . . we'll have to add that functionality and the appropriate objects.

Facilitator: Does that also add some constraints?

Jamie: It does, both technical and legal.

Production rep: Meaning?

Jamie: We better make sure an outsider can't hack into the system, disarm it, and rob the place or worse. Heavy liability on our part.

Doug: Very true.

Marketing: But we still need that . . . just be sure to stop an outsider from getting in.

Ed: That's easier said than done and . . .

Facilitator (interrupting): I don't want to debate this issue now. Let's note it as an action item and proceed.

(Doug, serving as the recorder for the meeting, makes an appropriate note.)

Facilitator: I have a feeling there's still more to consider here.

(The group spends the next 20 minutes refining and expanding the details of the home security function.)

Observations

- Observations are time consuming, so they aren't suitable for every user or every task.
- Select important or high-risk tasks and multiple user classes for observations.
- Observations can be silent or interactive.
  - Silent observations are appropriate when busy users cannot be interrupted.
  - Interactive observations allow the BA to interrupt the user mid-task and ask a question.
- Document what you observe for further analysis after the session.

Questionnaires

- Questionnaires are a way to survey large groups of users to understand their needs.
- They are …
  - Inexpensive
  - a logical choice for eliciting information from large user populations
  - can be administered easily across geographical boundaries.
- The analyzed results of questionnaires can be used as an input to other elicitation techniques.

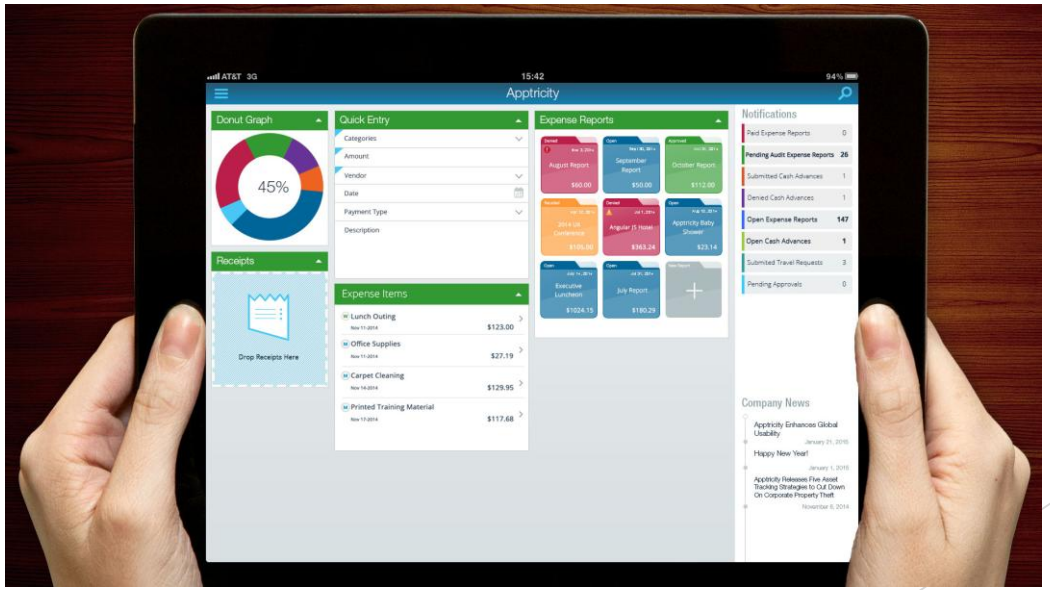
Questionnaires

Tips:

- Provide answer options that cover the full set of possible responses.
- Make answer choices both mutually exclusive and exhaustive.
- Don’t phrase a question in a way that implies a “correct” answer.
- If you use scales, use them consistently throughout the questionnaire.
- Choose wisely between closed questions and open-ended questions.
- Consider consulting with an expert.
- Always test a questionnaire before distributing it.
- Don’t ask too many questions or people won’t respond.

Interface analysis

- Interface analysis is an independent elicitation technique that entails examining the systems to which your system connects.
- System interface analysis reveals functional requirements regarding the exchange of data and services between systems.
- User interface (UI) analysis is an independent elicitation technique in which you study existing systems to discover user and functional requirements.



Questions

- You have been given the responsibility to elicit requirements from a customer who tells you he is too busy to meet with you. What should you do?

- Discuss some of the problem when requirements must be elicited from three or four different customers.

Other Techniques

- Scenario
- System interface analysis
- User interface analysis
- Document analysis
- …

In the next lecture…

- We will continue with the topic of requirements elicitation.
- Introduce other requirement elicitation techniques

Question session